

Anti Pairwise Comparison

Round 1:

Hover slide vs Snack n' Slide: Snack n' Slide worse (2 votes)

- Hover slide
 - Alters vending machine more
 - Replacing entire opening
 - Add slopes within the vending machine
 - Add bar outside of the vending machine
 - Requires significant maintenance (friction between the pulley and string, can cause snacks to get stuck on the way up)
 - Magnetic slider mechanism will be very hard to implement (high technical aspects of it)
- Snack n' Slide
 - Snack might not make it into the drawer
 - Still requires bending (Doesn't fully meet height requirements)
 - Requires a force still
 - User has to pull out, meaning they need to move back while squatting

Round 2:

Snack Chute vs Wave n' Take: Snack chute worse

- Wave and take
 - Significant changes - mechanical and electrical components
 - High risk of malfunctioning
 - Complex timing logistics and lots of edge cases
 - Difficult to implement
 - Increase complexity
 - Maintenance, complexity, what happens after failure
 - Not increasing height
- Snack chute
 - Snacks are light, it it can fall through flap then your even gonna get it
 - Snacks won't have enough force to open and will be stuck inside
 - Double collisions - products damages
 - A lot of changes implemented to vending machine - take stuff out

Round 3:

Hover slide vs Snack Sack: Snack Sack worse

- Snack Sack
 - Pouch is vulnerable to intentional damage
 - What if someone tries to purposely break the pouch
 - Or if the snack is very heavy (pouch is not strong enough)
 - Anti theft flap mechanism is not clear
 - Will require you to break the outside of the vending machine
 - Does not address the issue of retrieval height and also the force required to retrieve the snack
 - For the slope to work, the angle of the slope has to be sufficiently designed (has to be large? enough) → Snack needs to be able to slide down
- Hover Slide
 - Durability of the hover slide is concerning
 - There is a box outside of the vending machine that is attached by only one edge
 - Can be purposely destroyed
 - Can slide off
 - High chance of error (the box can be slid anywhere → not necessarily to the very left side → snack can get stuck within the vending machine → user does not get their snack)
 - Slope might not be steep enough to ensure that the snack will reach the bin
 - Requires significant alteration and requires high technology

Round 4:

Snack Chute vs Snack n' Slide: Snack n' Slide worse (unanimous)

- Snack n' Slide
 - More things that can go wrong
 - Pulling out motion can be odd? / not ergonomic for user
 - Has a worse anti-theft
 - Unless there is some miraculous other way to do anti-theft
 - Requires extra modification to the machine
- Snack Chute
 - Does not address the issue of bending
 - Only raises by 10 cm
 - Anti theft for the snack chute has not been tested — impossible to say if it actually works or not
 - Someone who is really strong can just punch their way up or use a tool to push it up

Round 5:

Wave n' Take vs Snack Sack: Wave n' Take worse

- Snack Sack
 - The force required to retrieve the snack is still huge compared to sensor
 - Still vulnerable to intentional damage
 - Design can be easily destroyed
- Wave n' Take
 - Doesn't increase retrieval height
 - Requires significantly more changes
 - Electrical, mechanical
 - Much more tedious to implement
 - Doesn't meet a lot of ECs like the height one (bending still required)

Round 6:

Hover Slide vs Snack Chute: Hover Slide worse

- Hover Slide
 - One of the main requirements is the force required - pulley may fail and you will need to put in a lot of force
- Snack Chute
 - Anti theft does not work as well
 - Bending down is required

Round 7:

Snack n' Slide vs Wave n' Take: Snack n' Slide worse

- Wave n' take:
 - Implementing a sensor requires significant structural changes
 - A lot of edge cases (how long will it take to open?)
 - Anti theft function doesn't change, down the line flap could start getting stuck again in the same manner
 - High chance of malfunction → Could be triggered too early, too late etc.
- Snack n' slide:
 - Requires more force
 - Over a long period of time
 - Does not meet anti-theft
 - Difficult to implement
 - Need to put in drawer, don't know dimensions
 - Drawer opening doesn't cover the whole bottom of the vending machine, what if the snack falls outside the drawer

Round 8:

Snack Chute vs Snack Sack: Snack Sack worse

- Snack sack
 - Pouch is vulnerable to intentional damage
 - What if someone tries to purposely break the pouch
 - Or if the snack is very heavy (pouch is not strong enough)
 - Anti theft flap mechanism is not clear
 - Will require you to break the outside of the vending machine
 - Does not address the issue of retrieval height and also the force required to retrieve the snack
 - For the slope to work, the angle of the slope has to be sufficiently designed (has to be large? enough) → Snack needs to be able to slide down enough
- Snack chute
 - Snack might not fall through the flap - possibly too light
 - Snacks can pile up
 - User might potentially not be getting their snack
 - Chances a lot of the original vending machine - take the base of the vending machine apart
 - Hard for first year Eng sci students to implement

Round 9:

Hover Slide vs Wave n' Take: Hover Slide worse

- Hover Slide
 - Doesn't satisfy force requirements (don't know if it satisfies)
 - Pulley slide mechanism has friction
 - Can malfunction in many places / many chances of failure
 - Box could fall off
 - High maintenance required for this design
- Wave n' Take
 - In the case of failure, it's impossible to retrieve this snack
 - Many edge cases / factor to consider
 - Snack falls on top of the flap
 - Sensor not activated
 - Increases complexity of vending machine

Round 10:

Snack Sack vs Snack n' Slide: Snack n' Slide worse

- Snack n' Slide:
 - Less intuitive than snack sack
 - Many components that can go wrong
 - Lots of alterations to the vending machine
 - Lifts up the snack but in a very complicated way versus the snack sack
- Snack sack
 - Anti theft limits the maximum opening of the flap if we don't figure out how to modify it according to the original vending machine
 - Would have to redesign the original vending machine